

| (a) (Sup,Sum <sup>+</sup> ) emptiness |     |     |     |     |     |     |     |     | (b) (LimSupAvg,Sum <sup>+</sup> ) emptiness |     |     |     |     |     |     |     |     |  | (c) (Sup,Sum <sup>B</sup> ) universality |     |     |     |     |     |     |     |     |
|---------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|---------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|--|------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|
| $n \backslash k$                      | 4   | 8   | 16  | 32  | 64  | 128 | 256 | 512 | $n \backslash k$                            | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  |  | $n \backslash k$                         | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   |
| 4                                     | .00 | .00 | .01 | .01 | .06 | .32 | 1.5 | 6.3 | 3                                           | .00 | .00 | .00 | .01 | .02 | .05 | .13 | .31 |  | 2                                        | .00 | .00 | .01 | .04 | .14 | .42 | 1.2 | 3.6 |
| 8                                     | –   | .00 | .01 | .03 | .16 | .76 | 3.3 | 14  | 4                                           | –   | .00 | .00 | .01 | .07 | .28 | 1.1 | 3.2 |  | 3                                        | –   | .00 | .02 | .08 | .40 | 2.3 | 12  | 44  |
| 16                                    | –   | –   | .01 | .07 | .39 | 1.7 | 7.3 | 32  | 5                                           | –   | –   | .00 | .03 | .21 | 1.3 | 5.5 | 20  |  | 4                                        | –   | –   | .02 | .12 | 1.1 | 12  | 159 | t   |
| 32                                    | –   | –   | –   | .13 | .85 | 3.8 | 17  | 72  | 6                                           | –   | –   | –   | .03 | .29 | 2.8 | 18  | 93  |  | 5                                        | –   | –   | –   | .14 | 2.0 | 51  | t   | t   |
| 64                                    | –   | –   | –   | –   | 1.3 | 7.8 | 35  | m   | 7                                           | –   | –   | –   | –   | .33 | 4.3 | 39  | 284 |  | 6                                        | –   | –   | –   | –   | 2.4 | 136 | t   | m   |
| 128                                   | –   | –   | –   | –   | –   | 12  | 71  | m   | 8                                           | –   | –   | –   | –   | –   | 4.7 | 58  | m   |  | 7                                        | –   | –   | –   | –   | –   | 177 | t   | m   |
| 256                                   | –   | –   | –   | –   | –   | –   | m   | m   | 9                                           | –   | –   | –   | –   | –   | –   | 63  | m   |  | 8                                        | –   | –   | –   | –   | –   | –   | t   | m   |
| 512                                   | –   | –   | –   | –   | –   | –   | –   | m   | 10                                          | –   | –   | –   | –   | –   | –   | –   | m   |  | 9                                        | –   | –   | –   | –   | –   | –   | –   | m   |

Table 1: Response time benchmarks: runtime in seconds, m = out of memory (30 GB), t = out of time (300 s).

| (a) (Sup,Max) emptiness |     |     |     |     |     |     | (b) (LimSupAvg,Max) emptiness |     |     |     |     |
|-------------------------|-----|-----|-----|-----|-----|-----|-------------------------------|-----|-----|-----|-----|
| $n \backslash k$        | 1   | 2   | 3   | 4   | 5   | 6   | $n \backslash k$              | 1   | 2   | 3   | 4   |
| 1                       | .00 | .00 | .00 | .00 | .00 | .00 | 1                             | .00 | .00 | .00 | .00 |
| 2                       | .00 | .00 | .01 | .06 | .37 | 1.9 | 2                             | .00 | .03 | 1.1 | 62  |
| 3                       | .01 | .10 | 1.1 | 11  | 103 | m   | 3                             | .12 | 48  | t   | t   |
| 4                       | .12 | 2.4 | 43  | m   | m   | m   | 4                             | 20  | t   | t   | m   |
| 5                       | 1.3 | 38  | m   | m   | m   | m   |                               |     |     |     |     |
| 6                       | 9.9 | m   | m   | m   | m   | m   |                               |     |     |     |     |

Table 2: Resource consumption benchmarks: runtime in seconds, m = out of memory (30 GB), t = out of time (300 s).